



AI4ALL Ignite Syllabus

AI4ALL Ignite is a transformative 2-semester accelerator designed to nurture the next generation of responsible AI Changemakers. Our program focuses on three key components:

- **Semester 1**
 - **Portfolio Project:** A 13-week immersive journey involving Technical Workshops, Responsible AI Discussions, group work, and mentorship from industry experts
 - **Student Symposium:** Students showcase their projects to industry sponsors, create networking opportunities, and have opportunities to learn about how different companies and organizations are moving the world of AI forward
- **Semester 2**
 - **Career Readiness:** An invigorating 7-week preparation for the internship application season, including Technical Interview Practice

This comprehensive accelerator equips students with technical and career preparation skills and the ethical grounding to stand out in the dynamic field of AI and contribute positively to its future.



- **Commitment:** Our staff is dedicated to each student's success, providing personalized guidance and support throughout the program.
- **Advocacy:** Our staff actively champions students' growth and potential, ensuring they have the resources and opportunities to excel.
- **Respect:** Our staff treats students with respect and values their contributions to the learning community.
- **Empowerment:** Our staff empowers students by fostering a supportive and inclusive learning environment, encouraging them to take ownership of their learning journey and career aspirations.



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Program Overview

Welcome to the AI4ALL Ignite Accelerator, a transformative 2-semester program where students will embark on a journey into the dynamic world of AI. Throughout this adventure, students will gain hands-on experience, connect with industry leaders, and develop a strong ethical foundation. Get ready to explore a path that will equip students with the skills and knowledge to make a real impact in the field of responsible AI.

Key Components

The AI4ALL Ignite accelerator has three key components, each of which is required.

1. Portfolio Project

In this exciting 13-week component, students will form groups to embark on an in-depth exploration of AI and machine learning. Guided by industry experts, students will explore a wide array of AI/ML topics and work collaboratively to create a remarkable portfolio project. This isn't just coursework; it's an opportunity for students to gain valuable technical skills and insights, fostering their growth and expertise in AI and machine learning.

At the core of this component is the Essential Question that underpins responsible AI:

★ Essential Question

How can the development processes of responsible AI/ML solutions address and mitigate the technology's negative impacts while strengthening its positive effects?

This question will guide students' journey as they shape the future of AI to make it more ethical and impactful.

2. Student Symposium

The Student Symposium (date and time TBD) marks the culmination of students' technical training. It's an opportunity for students to showcase their dedication, talent, and innovation to an esteemed audience of industry sponsors and peers. This event provides students with a platform to present the outcomes of the Portfolio Project and establish invaluable connections with Mentors and potential employers. It's time for students to shine, where their commitment to responsible AI is celebrated, leaving a lasting impression as they chart their path to an exciting AI career.



3. Career Readiness

After completing the Portfolio Project, students embark on an invigorating 7-week journey, tailored to equip them for internship application season. The Career Readiness component has two main focuses: Internship Application Preparation and Technical Interview Practice.

(i) Internship Application Preparation

This focus encompasses a Resume & LinkedIn Peer Reviews and Mentor Feedback Sessions. Students will also have the opportunity to network with potential employers.

(ii) Technical Interview Practice

Technical interviews can be challenging, but this focus ensures students are well-prepared to ace them. Students will engage in mock interviews and receive constructive feedback to improve their interview skills. We'll guide students on how to effectively communicate their technical knowledge and problem-solving abilities, enhancing their confidence when facing AI-related technical interviews.

Students will be well-prepared to navigate the professional landscape in AI and secure valuable internships and entry-level roles. This is a student's chance to develop the skills required to launch a successful career in the field of responsible AI.

Changemakers Connect

In addition to the three key components, students who have completed the Portfolio Project will be added to the Changemakers Connect platform. Through this platform, AI4ALL will provide listings of internships and entry-level roles and opportunities for students to build a community with one another.



Program Schedule

1. Portfolio Project

Program Kickoff	• Welcome message • Intro to Ed STEM (our LMS)
Week	Activity
01	Workshop: Project Ideation & Ethics • SMART Goals & Agile Scaffolding • Responsible AI Frameworks • Project Feasibility & Impact
02	Workshop: Data Sourcing & Privacy • Dataset Identification & Procurement • Data Privacy Laws & Regulations • Ethical Sourcing & Transparency Group Sign-ups
03	Workshop: Data Exploration & Cleaning • Exploratory Data Analysis (EDA) • Handling Missing Data & Outliers • Identifying & Mitigating Data Bias
04	Workshop: Advanced Model Specialization • Natural Language Processing with BERT • Computer Vision with CNNs & YOLO • K-Means Clustering for Unsupervised Learning Mentor Panel: Responsible AI
05	Workshop: Statistical Foundations & Visualization • Descriptive & Inferential Statistics • Data Visualization Best Practices • Communicating Data Insights
06	Workshop: Supervised Learning Architectures • Multiple Linear Regression • Logistic Regression • Decision Trees & Random Forests Mentor Panel: Machine Learning Models
07	Proposal Showcase: Project Proposal Presentations



08	Workshop: Model Validation & Interpretability ● Performance Metrics & Confusion Matrices ● K-Fold Cross Validation ● Explainable AI (XAI) & Auditing Mentor Panel: Student Skills
09	Workshop: MLOps & Deployment ● Environment Management with Docker ● API Integration & Cloud Computation ● Model Monitoring & Governance
10	Workshop: Results & Documentation ● Technical Result Synthesis ● Professional Project Documentation ● Communicating Technical Complexity
11	Workshop: Presentation Refinement & Showcase ● Drafting Narratives & Visuals ● Peer Feedback & Iterative Improvement ● Stakeholder Communication Strategies
12	Final Showcase: Final Project Presentations Career Readiness Sign-ups & Mid-Program Survey (15 min)
13	Workshop: GitHub Pages ● Setup Dedicated Repo ● Stylize Main Results ● Showcase on Resume



2. Student Symposium

Duration	Activity
5 minutes	Opening Remarks
10 minutes	Keynote
50 minutes	Student Presentations 1
5 minutes	Break
50 minutes	Student Presentations 2
25 minutes	Industry Professional Panel
5 minutes	Closing Remarks & Event Survey



3. Career Readiness

Week	Topic
1	Program Expectations & Assignments Overview - Walk through all assignments and expectations for students to complete the program officially Resume/LinkedIn Peer Review - Meeting with 1-2 peers
2	Networking and Applying - Create a plan for where to apply - Discuss best practices for networking and draft messages to connect Industry Mentor Resume Review - Each volunteer takes on 3 students per review for 2 reviews
3	Peer Behavioral Interview & Elevator Pitch Practice - Apply best practices to talk about your AI project in an interview - Practice additional behavioral interview questions with peers Final Resume Due
4	STUDENT SYMPOSIUM!
5	Technical Interview Practice I: Practice Questions - Students engage in technical interview/assessment questions - Form small groups, ask questions to a volunteer who curates the questions, and then the volunteer explains the answers to the large group
6	Technical Interview Practice II: Practice Questions (Cont.) Mock Technical Interview Assignment Due Students will complete their mock interview assignment on their own time during this week
7	Resources & Planning for Growth Share additional technical interview resources for continued preparation



Academic Requirements

Prerequisites

To ensure a smooth and effective learning experience, the accelerator operates under the assumption that students meet the following academic and technical criteria:

- Enrolled in a major/minor related to Computer Science, which may include disciplines such as computer science, data science, electrical engineering, or mathematics
- Proficiency in basic programming in Python, equivalent to the knowledge gained from a course like CS 1
- A foundational understanding of Probability and Statistics
- Competency in Linear Algebra
- A basic understanding of Python libraries, GitHub Terminology, and AI applications

These criteria are designed to help students excel in the program and fully engage with the content, projects, and discussions related to responsible AI.

Deliverables

At AI4ALL Ignite, we champion the transformative power of hands-on experience. We believe that learning truly comes to life when theory meets practice. Throughout our program, students will have access to essential takeaways, including a portfolio project, a polished GitHub Page showcase, and a My AI Roadmap for their continued journey.

Portfolio Project

The Portfolio Project is their chance to demonstrate the depth of their learning and the breadth of their capabilities. This project serves as the culmination of their 13-week journey. It's not just an assignment; it's a showcase of their potential in the world of responsible AI.

Project Proposal

A 1 to 2-page Project Proposal will be prepared, outlining the grand vision aimed to be achieved in the final presentation and project. This step provides students with the opportunity to set the stage for greatness.

Project Proposal Presentation

During a synchronous event, a summary of the Project Proposal will be presented to peers. This isn't just about showing what is planned; it's about convincing the audience that the project is groundbreaking.



Final Presentation

The Final Presentation is your opportunity to showcase your hard work and accomplishments within the classroom. As you present your final project, you'll share your knowledge, skills, and insights with peers and instructors, setting the stage for future learning and development. This presentation is a critical step toward mastering your subject and preparing for the next phase of your academic or professional journey.

GitHub Page

The Student Symposium is where you step into the spotlight beyond the classroom. With industry sponsors and key stakeholders in the audience - potential employers, internship providers, and graduate school representatives - this is your moment to make a lasting impression. Your GitHub Page showcases not just your work but your potential to impact the AI landscape, offering a powerful platform to connect with the broader AI community.

GitHub Repository

A GitHub repository will be created to showcase the project. This is not just a repository; it is a reflection of dedication and a resource for potential employers. It is where coding prowess, teamwork, and commitment to excellence are demonstrated.

Submitted Current Resume to the AI4ALL Resume Book

The AI4ALL Resume Book will be shared with partners who are interested in employing AI4ALL talent in internships and entry-level roles. Throughout the Career Readiness component of the program, students will take steps to improve their resumes and LinkedIn so that they can showcase their best selves in the AI4ALL Resume Book.

Internship Application/Networking Assignment

To help move students along in securing an internship, they will be required to submit proof 5 separate times that they have applied to an internship or entry-level job they are interested in and qualify for; Or they can submit proof of a networking attempt via email or on LinkedIn. This assignment is for students to help prioritize the opportunity search amidst their many responsibilities.

Mock Technical Interview

Amidst workshops to help prepare students for a technical interview, each student will be required to complete a mock technical interview to expose them to the technical interview experience. Demystifying this portion of the technical job application process will help them feel confident in taking these on in the future!



Rubrics

For a detailed breakdown of deliverable requirements, please refer to the [AI4ALL Ignite Rubrics](#).

Certificate of Completion

To receive a Certificate of Completion from AI4ALL Ignite, students must fulfill the following criteria:

- Attend a minimum of thirteen (13) cohort meetings throughout the program, including:
 - A minimum of eight (8) Portfolio Project meetings
 - A minimum of three (3) Panels or complete their make up assignments
 - A minimum of five (5) Career Readiness meetings
- Submit the Project Proposal and Final Presentations on time
- Actively participate in the Proposal Showcase
- Actively participate in the Final Showcase
- Complete all Group Work Activities
- Submit all Ed STEM assignments
- Submit the GitHub Repository
- Submitted Current Resume to the AI4ALL Resume Book
- Submit Identified Opportunities
- Complete the Mock Technical Interview



Program Structure

Stakeholders

AI4ALL Ignite (as a Whole)

Students: At the core of AI4ALL Ignite, our program's heartbeat, are the students. Our students are the reason we exist, and their role is nothing short of vital. As eager learners and future leaders in responsible AI, students represent the driving force behind our mission.

Sponsors: Supporters of AI4ALL with a special interest in seeing the success of our students.

Portfolio Project

Instructors: Instructors will present the Responsible AI Workshop synchronous sessions. They are responsible for delivering course content, facilitating discussions, and guiding the technical and ethical aspects of AI.

Student Coordinators & AI Project Leads: SCs and AIPLs will act as "project managers" during the Portfolio Project component. They will assist in organizing group activities, coordinating project-related tasks, and providing support to students during the accelerator. Additionally, SCs play a key role in ensuring the smooth functioning of the accelerator.

Mentors: Industry professionals will serve as Mentors by hosting impactful and relevant panel talks throughout the Portfolio Project. They will provide real-world insights and advice, helping students connect their learning to industry applications.

Career Readiness

Instructors: Instructors will present Internship Application Preparation and Technical Interview Practice sessions. They are responsible for delivering course content, facilitating discussions, and guiding students through their preparation and practice.

Student Coordinators: SCs will support the Career Readiness program administratively. They will also be present to support different volunteer-focused events.

Volunteers: Our dedicated volunteers offer their time, expertise, and passion to enhance the student experience. They may assist in various program activities and contribute to the overall success of AI4ALL Ignite. Their commitment is a testament to the collaborative and supportive community we've built.



Types of Meetings

Class Meetings

Class meetings will consist of a workshop and teamwork time amongst groups. During the teamwork time, students will be placed in a breakout room with their groups, and Student Coordinators will move between breakout rooms to check on their progress.

Students are expected to attend at least thirteen (13) of their class meetings. If a student cannot attend a meeting, they must let their group and Student Coordinator know as soon as possible. Students are expected to be on time for every meeting and stay for the duration of the meeting.

Mentor Panels

Schedule & Format

Mentor Panels are scheduled during Weeks 04, 06, and 08 of the Portfolio Project. They each consist of approximately 1 hour of discussion, and 45 minutes of networking (totalling 1:45).

Students are expected to attend all 3 Mentor Panels.

Expectations for Conduct & Communication

Mentors and students are expected to abide by the [AI4ALL Community Code of Conduct](#). If a Mentor or a student ever feels like an inappropriate interaction took place, reach out to programs@ai-4-all.org. Safety is our top priority.

Communication Norms

Students are expected to treat their Mentors with respect and learn from their expertise. Students should conduct themselves as if they are in a professional or academic environment. Students should refrain from using inappropriate language or references.

Additional Opportunities During Mentor Panels

These panel can also serve as a platform for broader learning and career exploration. When appropriate, sessions may include:

- Discussions about career paths and experiences in the AI field
- Guidance on post-degree opportunities, such as graduate school or certifications
- Insights into current industry trends and skill requirements
- Conversations about navigating technical careers and building professional networks
- Tips on preparing for technical interviews and developing marketable skills



Policies

These policies protect the interests of students, Instructors, Student Coordinators, AI Project Leads, Mentors, volunteers, and AI4ALL as a whole. In unforeseen circumstances not covered by this document, we will base decisions on individual case considerations.

AI4ALL Community Code of Conduct

As a member of the AI4ALL community, we uphold the following principles and standards of conduct to ensure a positive and respectful environment for all participants. By joining our community, students agree to adhere to these guidelines:

1. Be a Responsible Digital Citizen

- Act as good virtual citizens by demonstrating respect for the needs and experiences of all members of the AI4ALL community, including peers, Mentors, staff, and fellow participants
- Demonstrate accountability and take responsibility for actions consistently
- Support the learning of fellow students and be understanding of their mistakes
- Respect each other's privacy and embrace diverse perspectives

2. Engage Actively & Thoughtfully

- Thoughtfully participate in AI4ALL programs by asking questions and sharing ideas
- Practice active listening during virtual interactions

3. Maintain Truthfulness & Respect

- Use appropriate language and methods of expression in all personal and academic interactions
- Refrain from using obscene, derogatory, or abusive language

4. Uphold Academic Integrity

- Maintain high standards of academic integrity by properly attributing work used or citing its respective owner(s)
- Avoid plagiarism and uphold the value of originality in academic pursuits

5. Promote Inclusivity & Equity

- Do not engage in or support any form of discrimination, including but not limited to discrimination based on sex, race, color, national or ethnic origin, age, religion, sexual orientation, gender identity, gender expression, status as a veteran, or disability



6. Respect Virtual Assets

- Respect and take care of AI4ALL and program participants' virtual assets, including software, digital resources, online spaces, and personal data; for example:
 - Protect sensitive information
 - Maintain strong cybersecurity practices
 - Be ethical and considerate in all online interactions

7. Reject Violence and Harassment

- Do not commit or threaten acts of violence against people or property
- Do not participate in any harassment or bullying that fosters an intimidating or offensive virtual environment

8. Avoid Disruptive or Unsafe Behavior

- Do not engage in disruptive or unsafe behavior or communication that interferes with the activities of fellow students, volunteers, or AI4ALL staff

9. Prioritize Well-being

- Prioritize the care of physical, mental, and emotional health
- Communicate needs openly with AI4ALL to ensure we can support well-being effectively

10. Compliance & Accountability

- Understand and follow this code of conduct and any extra guidelines given by AI4ALL staff throughout the program
- Recognize that AI4ALL reserves the right to remove participants from the program in case of code of conduct violations, behavior that jeopardizes safety or success, or actions beyond our organizational scope
- Refer to the [Attendance](#), [Withdrawal](#), and [Removal](#) Policies for more information

Attendance Policy

AI4ALL recognizes that active participation and regular attendance in program activities, including synchronous sessions such as workshops, launches, and presentations, are essential for student success.

Specific activities related to AI4ALL Ignite include:

- Enrolling in our learning management system (LMS)
- Registering for Changemakers Connect (Community platform)
- Attending workshops/program sessions, program launches, and presentations
- Attending group meetings and Biweekly Mentor Check-ins
- Submitting program assignments and deliverables



AI4ALL Ignite requires attendance as part of the protocol to receive Certificates of Completion. In unforeseen circumstances not covered by this document, we will base decisions on individual case considerations.

Students must attend the following for each portion of the program:

Portfolio Project

- The Program Kickoff event (Before Program Launch)
- The Program Launch (Week 1 of the program)
- The Proposal Presentations
- A minimum of eight (8) synchronous sessions in the Portfolio Project component
- Attend all three (3) Panels

Student Symposium

- The [Student Symposium](#) synchronous event

Career Readiness

- Industry Mentor Resume feedback session
- A minimum of five (5) synchronous sessions in the Career Readiness component
- At least one Opportunity Chat time slot

Approved Absences

To qualify for an approved absence, the following conditions must be met:

- A written request, along with any supporting documentation (e.g., doctor's note), must be submitted to the Instructor or Student Coordinator
- The request must receive written approval from the Instructor or Student Coordinator
- The student must submit the relevant makeup assignment for the missed session within the agreed-upon timeframe

Make-up Assignments

Students who are unable to attend a required synchronous session AND *meet the criteria for an [approved absence](#)* will be provided access to a makeup assignment for the missed session.



Withdrawal Policy

Types of Withdrawals

Official Withdrawal

An official withdrawal initiated by the student during the official withdrawal period is non-punitive. See the [How to Withdraw from the Accelerator](#) section of this document for the steps to withdraw from AI4ALL Ignite.

Administrative Withdrawal

AI4ALL may administratively withdraw a student who ceases to participate in program activities for more than ten days without notifying their Instructor. The possibility of enrolling in future AI4ALL programming will be evaluated in these cases.

Non-attendance Withdrawal

This withdrawal is assigned if a student did not actively participate in any program activities during the first seven (7) days after the launch and remained unresponsive to communication during the subsequent three (3) days. The eligibility for future AI4ALL programming will be determined on a case-by-case basis.

Note: Students who withdraw from the program, regardless of the type of withdrawal, are ineligible to receive Certificates of Completion.

Official Withdrawal Period

The official withdrawal period extends for the first twenty-one (21) days following the launch date of AI4ALL Ignite. After this deadline, withdrawals will be considered only for compelling reasons, such as a medical emergency, supported by adequate documentation from a healthcare provider, etc.

How to Withdraw from the Program

1. Email the Instructor to inform them of the decision to withdraw
2. Email the project team to notify them of the withdrawal decision
3. Optionally, complete an exit survey



Removal Policy

AI4ALL reserves the right to remove a student at any point from our accelerators if the student violates the [AI4ALL Community Code of Conduct](#) or disrupts program activities negatively affecting other students, Mentors, volunteers, or AI4ALL staff. Immediate removal or temporary suspension may occur based on the nature of the incident. A student who is immediately removed may not enroll in future AI4ALL programs.

Temporary Suspension

- **Disruptive Behavior or Communication:** A student may be temporarily suspended if they engage in disruptive behavior or communication that raises concerns or detracts from program activities

A student will receive an email from programs@ai-4-all.org letting them know if they are temporarily suspended, and the following steps will take place:

- AI4ALL staff will promptly reach out to the student's Instructor and teammates to inform them of the student's temporary withdrawal
 - The communication will not disclose the specific reason for the student's withdrawal to protect their privacy
- After a staff review, a student may be fully reinstated, conditionally reinstated, or removed from the program
 - **Full Reinstatement:** If the investigation determines that the student's behavior does not warrant further action, the student will be fully reinstated and allowed to rejoin the program without any conditions
 - **Conditional Reinstatement:** If the circumstances warrant reinstatement but with certain conditions, AI4ALL staff will specify these conditions and establish a trial period to ensure compliance
 - Violation of the specified conditions may result in the student's immediate removal from the program
 - **Removal:** If it is determined that the student's behavior cannot be adequately addressed within the program, they will be permanently removed. In such cases, AI4ALL will make individual determinations regarding the student's eligibility to participate in future semesters of programming

Immediate Removal

- **Physical or Psychological Harm:** If a student causes or intends physical or psychological harm to other students, volunteers, or AI4ALL staff, immediate removal from the program will be enforced
- **Plagiarism:** If a student is found to have plagiarized an assignment, either in part or in full, they will be immediately removed from the program
 - Plagiarism is considered a serious breach of academic integrity



Note: This policy serves as a framework for maintaining a conducive learning environment within AI4ALL Ignite. All participants need to adhere to these guidelines to foster a respectful and productive community.

Camera Policy

Instruction takes place in a virtual environment where students are expected to show that they are engaged and participating by having their **cameras on during synchronous meetings**, especially in breakout sessions/small group discussions. Students should try their best to be in a location where they can have their cameras on while on calls for our programs. They can also take advantage of available virtual backgrounds. If they are unfamiliar with this feature, they can learn more about it [here for Zoom](#) and [here for Google Meet](#) (requires using the Chrome browser). If using their camera is not possible on a particular occasion, they should let their Instructor and Student Coordinator know, and try to be able to have their camera on next time. Students should also participate fully in the breakout rooms by unmuting themselves. We encourage students to use the chat feature when they are in the main room.

Peer Feedback Protocol

Throughout the accelerator, students will provide feedback to their peers on various assignments, including their project proposals and final presentations. Feedback is most helpful when it is constructive and actionable. One way to do so is to use the C.O.I.N. formula: Context, Observation, Impact, and Next.

CONTEXT Define the situation <i>"During your presentation this morning..."</i>	OBSERVATION Describe the behavior <i>...I noticed that you were focused on the presentation screen..."</i>
IMPACT Clarify the impact <i>...It made me feel I didn't have a chance to ask questions or get clarity on parts I didn't understand..."</i>	NEXT Make a suggestion <i>...Facing your audience will allow you to gauge when to elaborate and remind you to stop for questions."</i>

The **C.O.I.N. Conversation Model**, adapted from Anna Carroll's *The Feedback Imperative*
<https://www.mindtools.com/a94k5vp/the-coin-conversation-model>